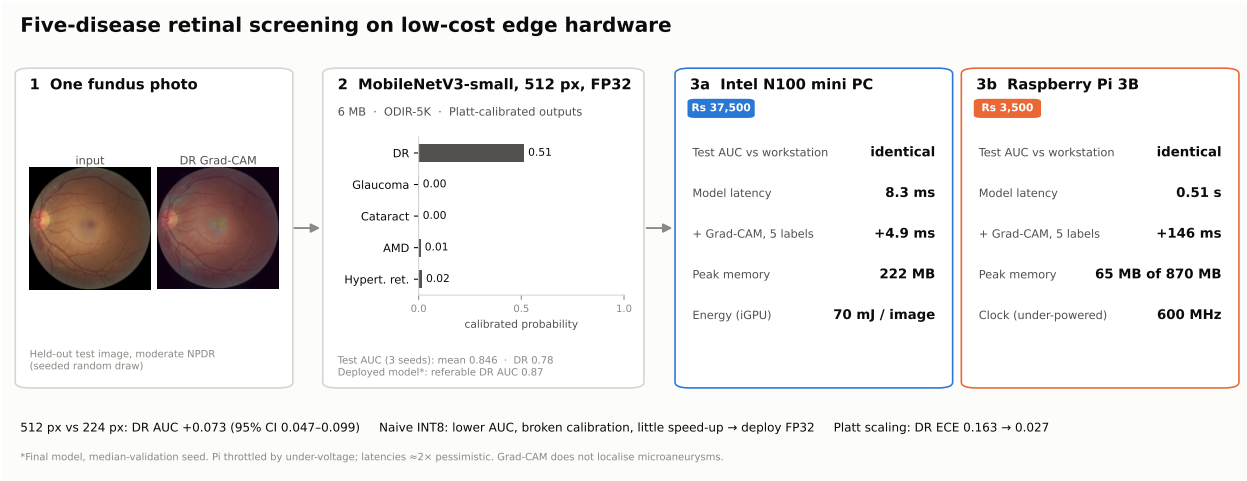


Graphical Abstract

Multi-disease retinal screening on low-cost edge hardware: accuracy, latency and quantization trade-offs of a lightweight multi-label network on a mini PC and a Raspberry Pi

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Highlights

Multi-disease retinal screening on low-cost edge hardware: accuracy, latency and quantization trade-offs of a lightweight multi-label network on a mini PC and a Raspberry Pi

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- One lightweight network screens five retinal diseases from a fundus photo
- FP32 export to an Intel N100 and a Raspberry Pi 3B is lossless
- 512-pixel input raises held-out DR AUC by 0.07; per-eye labels do not
- Naive INT8 quantization costs accuracy and breaks calibration
- Temperature scaling fails under class weighting; Platt scaling works

Multi-disease retinal screening on low-cost edge hardware: accuracy, latency and quantization trade-offs of a lightweight multi-label network on a mini PC and a Raspberry Pi

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ABSTRACT

Background: Diabetic retinopathy (DR), glaucoma, cataract and age-related macular degeneration cause most avoidable vision loss, but specialists and computing infrastructure are scarce where screening is most needed. **Methods:** A single MobileNetV3-small network screens five retinal conditions from one fundus photograph. It was trained on ODIR-5K and evaluated on a patient-disjoint held-out test set over three seeds, then deployed with ONNX Runtime, OpenVINO and TensorFlow Lite on an Intel N100 mini PC and a Raspberry Pi 3B. We measured export fidelity, latency, memory, energy, INT8 quantization, calibration and on-device Grad-CAM. **Results:** Raising input resolution from 224×224 to 512×512 improved mean test AUC from 0.800 to 0.846 and DR AUC by 0.073 (95% CI 0.047–0.099). Per-eye DR labels added no measurable gain. Floating-point deployment was lossless: test AUCs matched the reference to four decimals on both devices. The model needed 8 ms and 222 MB on the N100, and 0.51 s and 65 MB on the Pi. Naive INT8 quantization cost up to 0.23 AUC and broke calibration for little speed gain. Class weighting made outputs over-confident. Temperature scaling failed to correct this, while Platt scaling reduced DR calibration error from 0.163 to 0.027. On-device Grad-CAM matched the reference, passed randomisation sanity checks and added 9–27% latency on the Pi, but never highlighted microaneurysms. **Conclusion:** A floating-point model using under 100 MB on a low-cost single-board computer matches its workstation accuracy for five-disease screening, provided calibration includes a bias term and quantization is avoided.

1. Introduction

At least 2.2 billion people live with a vision impairment, and in at least 1 billion of them it could have been prevented or has yet to be addressed [1]. Cataract, glaucoma, age-related macular degeneration (AMD) and diabetic retinopathy (DR) are among the leading causes of blindness [2]. About 90% of people with distance vision impairment live in low- and middle-income countries [3], which also have the fewest eye specialists: the mean density of ophthalmologists is 3.7 per million inhabitants in low-income countries, against 76.2 per million in high-income countries [4]. All four conditions, and hypertensive retinopathy, leave visible signs in a colour fundus photograph, which makes automated fundus screening an attractive way to extend scarce specialist capacity.

Deep convolutional networks reach specialist-level accuracy for single diseases on large curated datasets: an AUC of 0.99 for referable DR [5], 0.986 for glaucomatous optic neuropathy [6] and 0.94–0.96 for AMD [7]. Three gaps limit their use in low-resource screening. First, most systems target a single disease, while multi-disease systems rely on hundreds of thousands of images and server-class models [8, 9]. Second, performance in deployment is less reliable than on curated data: seven commercial DR algorithms evaluated on the same real-world teleretinal images reached sensitivities between 50.98% and 85.90% [10], and image quality, workflow and connectivity constrained a deployed system in Thai clinics [11]. Third, most systems assume a workstation GPU or a cloud connection. Offline on-device screening has been validated clinically, but only for proprietary single-disease DR software on smartphones [12, 13]. Academic edge deployments rarely report whether the deployed model still matches the trained one, how much memory it needs, or what quantization costs for each disease (Section 2).

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This paper measures how much multi-disease screening accuracy survives when a single lightweight network is deployed on two inexpensive devices: an Intel N100 mini PC and a Raspberry Pi 3B with less than 1 GB of usable RAM. The contributions are:

- (1) a multi-label MobileNetV3-small model that screens five retinal conditions (DR, glaucoma, cataract, AMD and hypertensive retinopathy) from a single fundus photograph, trained on ODIR-5K and evaluated on a patient-disjoint held-out test set with bootstrap confidence intervals;
- (2) an analysis of why DR performs poorly on ODIR-5K. Interference between the five disease heads is not the cause, and input resolution is: on the held-out test set, 512×512 input raises DR AUC by 0.05–0.07 over 224×224 . Per-eye DR labels, which looked helpful during development, gave no measurable gain on the test set;
- (3) a reproducible edge benchmark of the same model on both devices with ONNX Runtime, OpenVINO and TensorFlow Lite. It reports per-label AUC, deviation from the reference model, latency distribution, peak memory and, on the N100, energy per inference;
- (4) a negative result for naïve INT8 post-training quantization of this architecture: it costs accuracy and shifts per-image probabilities, without a meaningful speed or memory benefit on either device;
- (5) an on-device Grad-CAM path that needs no automatic differentiation on the device, evaluated with sanity checks, deletion/insertion faithfulness and localisation against expert lesion masks, together with its cost on both devices;
- (6) an assessment of confidence calibration on the held-out test set. Temperature scaling fails for this model, and Platt scaling works. The calibrated outputs survive FP32 deployment exactly, while full-integer INT8 breaks them.

2. Related work

2.1. Deep learning for retinal disease screening

Gulshan et al. trained a DR classifier on 128,175 images and reported AUCs of 0.991 and 0.990 for referable DR on two validation sets [5]. Ting et al. extended the approach to a multi-ethnic population and added glaucoma and AMD, using a separate model for each disease [8]. Single-disease systems for glaucoma [6] and AMD [7] report similar accuracy. In a prospective pivotal trial, the first autonomous DR system authorised by the FDA reached 87.2% sensitivity and 90.7% specificity [14], below the retrospective figures. Two causes of this gap recur in the literature. The first is image and label quality: adjudicated reference grades together with higher-resolution input raised the AUC for moderate-or-worse DR from 0.934 to 0.986, and missed microaneurysms were the most common grading error [15]. A reproduction of the original DR system on public data with single-grader labels reached only 0.853 on Messidor-2 [16]. The second cause is the deployment setting itself [10, 11].

2.2. Multi-label fundus classification

ODIR-5K provides colour fundus photographs of both eyes for 5,000 patients, with eight diagnostic categories assigned per patient [17, 18]. Work on ODIR has mostly maximised accuracy with transfer learning, ensembles or bilateral fusion [19–24]. None of these reports results on low-cost hardware. RFMiD adds 3,200 images from three cameras with 46 conditions [25]. Large multi-label platforms reach an AUC of 0.998 across 39 conditions, but they are trained on about 250,000 images [9].

2.3. Lightweight networks and quantization

MobileNetV3 combines depthwise-separable convolutions, squeeze-and-excitation (SE) blocks and hard-swish activations, and was designed for mobile CPUs [26–28]. Post-training INT8 quantization usually stays within about 2% of floating-point accuracy when weights are quantized per channel [29, 30], and integer-only inference underpins TensorFlow Lite [31]. Depthwise-separable networks are a known exception: MobileNets often lose substantial accuracy under post-training quantization, because quantization error accumulates through the depthwise layers [32, 33]. Data-free equalisation and bias correction partly recover this loss [34].

2.4. Edge and offline retinal screening

Offline smartphone AI for referable DR has been validated in community screening by minimally trained health workers [12, 13]. Academic edge work mostly targets DR alone, on a Raspberry Pi 4 [35], a Raspberry Pi or Jetson Nano [36], or smartphones and microcontrollers [37, 38]. Two studies run multi-disease models on edge devices: ODIR classifiers on a Jetson Nano [39], and an eight-class EfficientNet-B0 with spatial attention on a Raspberry Pi 5, at about

5 s per image [40]. Energy per image has been measured for glaucoma classification on an Edge TPU and a Jetson [41]. Reports on quantization disagree. Some describe INT8 as nearly free [38, 42], while a four-class retinal MobileNet study found FP16 better than its integer variants [43].

2.5. Gap addressed by this work

None of these edge studies reports, on the target device, whether the deployed model matches the trained one (AUC and per-image output), together with peak memory and a per-disease account of quantization cost. None targets the least capable tier, a Raspberry Pi 3B with 1 GB of RAM and a 32-bit operating system. We provide these measurements for a five-condition model on two devices that differ by an order of magnitude in price and compute.

3. Materials and methods

3.1. Dataset and splits

We use ODIR-5K [17, 18], obtained as the widely used preprocessed public mirror, with images cropped and resized to 512×512 . It contains 6392 per-eye fundus images from 3358 patients, annotated with eight patient-level categories. We model five: DR (D), glaucoma (G), cataract (C), AMD (A) and hypertensive retinopathy (H). The normal and “other” categories are not modelled. Patient-level prevalence is 1105 (D), 206 (G), 208 (C), 163 (A) and 103 (H) of 3358 patients, so every label except DR is rare.

All splits are grouped by patient, so both eyes of a patient always fall in the same partition. A first grouped split (20%, seed 42) defines a validation set of 1270 images. This set was used during development to choose input resolution and DR label definition (Section 4.2), and in the final protocol only to select checkpoints and operating thresholds. From the remaining images, a second grouped split (20%, seed 2026) defines a held-out test set that no model was evaluated on until the final evaluation. The rest is used for training. The final partitions contain 4097 training images (2148 patients), 1270 validation images (672 patients) and 1025 test images (538 patients). The test set has 285 eyes with DR under per-eye labels and 62, 62, 42 and 34 positives for glaucoma, cataract, AMD and hypertensive retinopathy.

Per-eye DR labels. ODIR-5K assigns its labels per patient, so an eye without retinopathy inherits a positive DR flag when the fellow eye is affected. We derive per-eye DR labels from the left- and right-eye diagnostic keywords supplied with the dataset. Nine keywords denote DR (mild, moderate and severe non-proliferative DR, proliferative DR, and generic or suspected DR). Hypertensive and myopic retinopathy are excluded. These keywords reproduce the patient-level flag exactly for all 1105 positive patients. Of the 2123 images whose patient is DR-positive, 401 (18.9%) carry no DR keyword for that eye; 250 of them are annotated as “normal fundus”. The keywords also provide DR severity, which we use for a referable-DR analysis (moderate non-proliferative DR or worse).

3.2. Model and training

The classifier is MobileNetV3-small [26] (mobilenetv3_small_100 from timm [44], pretrained on ImageNet [45]) with a five-way sigmoid output. Training uses AdamW (learning rate 3×10^{-4} , weight decay 10^{-4}) for 25 epochs, with a two-epoch linear warm-up, cosine decay and batch size 32. To handle class imbalance we use binary cross-entropy with a positive weight per label equal to the ratio of negatives to positives in the training set, clipped to [1, 20]. Training augmentation consists of random resized crops (scale 0.8–1.0), horizontal flips, rotations of up to $\pm 20^\circ$ and colour jitter (brightness and contrast 0.2, saturation 0.1). Evaluation images are only resized and normalised with ImageNet statistics. We keep the checkpoint with the highest mean validation AUC.

We train three configurations with three random seeds each: (A) 512×512 input with per-eye DR labels; (B) 224×224 with the original patient-level labels, the baseline; and (C) 512×512 with patient-level labels, which separates the effect of resolution from that of the label definition. For deployment we take the configuration-A seed with the median validation mean AUC, chosen before any test evaluation.

3.3. Edge deployment pipeline

The trained network is exported to ONNX (opset 17) for the N100 and to TensorFlow Lite for the Raspberry Pi. On each device we re-implement preprocessing (bilinear resize and ImageNet normalisation) to match the training pipeline exactly. We verify export fidelity on every evaluation image by comparing per-image output probabilities with those of the PyTorch model on the workstation CPU. Runtimes are ONNX Runtime [46] and OpenVINO [47] on the N100, and the TensorFlow Lite (LiteRT) interpreter [48] on the Pi.

Quantization. For the N100 we apply OpenVINO/NNCF [49] post-training quantization (mixed preset) calibrated on 300 images drawn from the *training* split. For the Pi we evaluate TensorFlow Lite dynamic-range quantization (8-bit weights, float activations) and full-integer quantization with the same calibration images. No quantization-aware training or accuracy-aware tuning is applied; this is deliberately the lowest-effort path a deployer would try first.

3.4. Confidence calibration

Positive-class weighting in the loss inflates predicted probabilities for rare labels, so raw outputs are not expected to be calibrated. For each label we fit, by minimising the validation negative log-likelihood, either a single temperature, $\sigma(z/T)$ [50], or a Platt map with scale and bias, $\sigma(az + b)$ [51], where z is the output logit. A positive-class weight w in the loss shifts the optimal logit by about $\log w$. A bias term can undo that shift, but a temperature cannot. Both maps are monotonic, so they leave AUC unchanged, and on the device they cost one or two arithmetic operations per label. On the test set we report the expected calibration error (ECE) with 15 equal-width bins [50, 52], an equal-mass (adaptive) variant, the Brier score and the negative log-likelihood, each before and after scaling. We obtain confidence intervals for the change in ECE from 1000 bootstrap resamples of test patients. ECE estimated on resampled data is biased upward, so we do not report percentile intervals for ECE itself. We also apply the validation-fitted temperatures to the probabilities produced on each device, and compare the raw ECE of FP32 and INT8 variants of the development model.

3.5. Visual explanations

We compute Grad-CAM [53] on the last convolutional feature map for each of the five outputs. On the edge devices the network is split in two. The exported backbone returns the feature map, and a small NumPy implementation computes the classification head and the Grad-CAM gradient analytically from the exported head weights, so the device needs no automatic differentiation. We check that the maps match the PyTorch reference, and evaluate them in three ways. The first is the parameter-randomisation sanity check [54]. The second is deletion and insertion faithfulness on the held-out test set, against random and centre-bias baselines. The third is localisation against pixel-level DR lesion masks from IDRiD [55]. Deletion and insertion progressively remove (by blurring) or restore the most salient pixels and track the probability of the target label. We summarise each curve by its area, over 485 test image-label pairs with positive labels. IDRiD provides 81 images with expert masks for microaneurysms, haemorrhages, hard and soft exudates. We obtained it from a public mirror under its CC BY 4.0 licence, cropped and resized the images to 512×512 like ODIR, and applied the same transform to the masks. For localisation we report the pixel-level AUROC of the DR map against the lesion masks and the pointing game (whether the map maximum falls inside a lesion). Baselines are a random map, a centred Gaussian (centre bias) and the glaucoma map of the same image (wrong-class control). On the devices we time the backbone together with the NumPy head and Grad-CAM, with the numerical library pinned to one thread. With default threading, the overhead rose roughly sixteen-fold on the N100, and one run crashed on the Pi.

3.6. Hardware and benchmarking protocol

Table 1 summarises the two devices.

Table 1

Edge devices used for benchmarking.

	Intel N100 mini PC	Raspberry Pi 3B
CPU	4-core x86-64, up to 3.4 GHz	4-core ARM Cortex-A53 (ARMv7 OS), 1.2 GHz
Accelerator	Intel UHD iGPU (OpenVINO)	none
RAM	15 GB	870 MB usable
OS / Python	Debian 13, Python 3.13.5	Raspbian 11, Python 3.9.2
Runtimes	ONNX Runtime 1.30.0, OpenVINO 2026.4.1	TensorFlow Lite runtime 2.13.0
Price paid (INR)	Rs 37,500	Rs 3,500

Each configuration runs in its own process, so that peak resident memory (RSS) is attributable to it alone. We measure accuracy over all 1270 validation images at batch size 1. We measure model-only latency on one fixed preprocessed image after warm-up: 50 warm-up and 500 timed runs on the N100, and 20 warm-up and 200 timed runs

on the Pi. We report mean and 95th-percentile latency, along with the mean time for preprocessing (JPEG decoding and resizing). On the Pi we read the clock frequency and the firmware throttling flags before and after every configuration. On the N100 we measure package energy with Intel RAPL over sustained 60 s loops, three repeats per configuration with a 30 s cool-down, both for inference alone and end to end. An idle baseline is measured before and after each pass. The Pi has no on-board energy counter.

3.7. Evaluation metrics

We report the area under the ROC curve (AUC) for each label and its unweighted mean across the five labels. For export fidelity we report the maximum and mean absolute probability difference from the reference. Test AUCs are reported as the mean $\pm$ standard deviation over three seeds. The 95% confidence intervals come from 2000 bootstrap resamples of test patients, applied to the seed-averaged AUC, and paired resamples give intervals for the differences between configurations. For each label we choose a threshold on the validation set that gives 90% sensitivity, and report sensitivity and specificity on the test set at that fixed threshold. For DR we also report AUC by severity grade against eyes without DR, and for referable DR (moderate non-proliferative or worse against the rest).

4. Results

4.1. Held-out test performance

Table 2 reports the three configurations on the 1025-image held-out test set.

Table 2

Held-out test AUC (1025 images, 538 patients): mean $\pm$ s.d. over three seeds, with the 95% patient-bootstrap CI of the seed-averaged AUC in brackets. (A) 512 $\times$ 512, per-eye DR labels; (B) 224 $\times$ 224, patient-level DR labels; (C) 512 $\times$ 512, patient-level DR labels.

Label	A: 512, per-eye	B: 224, patient	C: 512, patient
DR (per-eye labels)	0.780 $\pm$ 0.032 [0.747, 0.812]	0.708 $\pm$ 0.028 [0.670, 0.742]	0.775 $\pm$ 0.031 [0.741, 0.806]
DR (patient labels)	0.752 $\pm$ 0.023 [0.716, 0.786]	0.698 $\pm$ 0.024 [0.661, 0.735]	0.755 $\pm$ 0.019 [0.720, 0.790]
Glaucoma	0.889 $\pm$ 0.012 [0.822, 0.939]	0.860 $\pm$ 0.004 [0.798, 0.908]	0.885 $\pm$ 0.008 [0.815, 0.937]
Cataract	0.918 $\pm$ 0.023 [0.873, 0.955]	0.905 $\pm$ 0.015 [0.850, 0.948]	0.920 $\pm$ 0.009 [0.872, 0.957]
AMD	0.852 $\pm$ 0.024 [0.765, 0.924]	0.841 $\pm$ 0.022 [0.735, 0.928]	0.878 $\pm$ 0.011 [0.798, 0.946]
Hypertensive ret.	0.791 $\pm$ 0.021 [0.703, 0.871]	0.694 $\pm$ 0.038 [0.609, 0.776]	0.798 $\pm$ 0.009 [0.705, 0.880]
Mean of five	0.846 $\pm$ 0.017	0.800 $\pm$ 0.018	0.847 $\pm$ 0.004

Higher resolution is the effect that holds on unseen data. Configuration A exceeds the 224 $\times$ 224 baseline B for DR by 0.073 (95% CI 0.047 to 0.099) under per-eye labels and by 0.053 (0.029 to 0.080) under patient-level labels. It also exceeds B for hypertensive retinopathy by 0.097 (0.028 to 0.164) and for glaucoma by 0.029 (0.004 to 0.053). Cataract and AMD do not differ significantly. Per-eye DR labels, by contrast, bring no measurable benefit at equal resolution: A and C differ in DR AUC by 0.005 (−0.014 to 0.024), and A is slightly worse for AMD (−0.027, −0.048 to −0.007). Seed-to-seed variation is substantial, with a standard deviation of 0.02–0.03 for DR.

The deployed model (configuration A, seed with the median validation score) reaches a test DR AUC of 0.809 under per-eye labels and 0.771 under patient-level labels. Its AUC for referable DR is 0.869. For DR graded against eyes without DR, its AUC is 0.667 for mild non-proliferative DR (86 eyes), 0.865 for moderate (156), 0.940 for severe (25) and 0.966 for proliferative DR (6). For the other labels it reaches 0.892 (glaucoma), 0.897 (cataract), 0.862 (AMD) and 0.813 (hypertensive retinopathy). This seed happens to be the strongest of the three on test DR, so the seed means in Table 2 are the fairer summary. With thresholds fixed on the validation set at 90% sensitivity, test sensitivity and specificity are 0.89/0.47 for DR, 0.89/0.67 for glaucoma, 0.81/0.82 for cataract, 0.81/0.75 for AMD and 1.00/0.29 for hypertensive retinopathy. At such operating points the model would refer many patients without the disease.

4.2. Development analysis: why DR underperforms

All numbers in this subsection are on the validation set used for development. They motivated the choice of resolution and label definition, which Section 4.1 then tests on unseen data. Of the two effects seen here, only resolution held up on the test set.

At 224×224 the joint model reached a DR AUC of 0.695, against 0.82 to 0.94 for the other diseases except hypertensive retinopathy. A single-task DR model trained with the same recipe reached 0.719. The difference of 0.025 has a paired bootstrap 95% CI of $[-0.003, 0.053]$, so interference between the five heads explains little of the gap. Table 3 varies input resolution and DR label definition.

Table 3

DR AUC on the development (validation) set by input resolution and training label. Two seeds per configuration where shown as “seed 1 / seed 2”. Mean AUC is over all five labels, with DR scored against patient-level labels.

Input	DR training label	DR AUC (patient)	DR AUC (per-eye)	Referable DR AUC	Mean AUC
224	patient	0.695 / 0.687	0.714 / 0.706	0.721 / 0.706	0.822 / 0.822
224	per-eye	0.700	0.717	0.721	0.827
384	patient	0.752 / 0.714	0.764 / 0.734	0.792 / 0.753	0.848 / 0.843
384	per-eye	0.734	0.763	0.793	0.843
512	patient	0.753 / 0.755	0.772 / 0.777	0.787 / 0.811	0.859 / 0.847
512	per-eye	0.769 / 0.783	0.803 / 0.813	0.841 / 0.849	0.856 / 0.861

Resolution is the main factor. At 512×512 , DR AUC rises by 0.06 to 0.10 over the 224×224 baseline, and the paired bootstrap interval excludes zero for all four 512 -pixel runs. Per-eye labels make no difference at 224×224 and appeared to add about 0.03 at 512×512 in both seed pairs. This gain did not replicate on the held-out test set (Section 4.1). This is consistent with lesion-scale evidence that small lesions such as microaneurysms need high resolution to be learned at all [15, 56]. Higher resolution did not harm the other labels: hypertensive retinopathy gained most, and glaucoma, cataract and AMD stayed within seed-to-seed variation. The remaining DR errors concentrate in mild non-proliferative DR. Against no-DR eyes the 512 -pixel per-eye models reached 0.70–0.73 AUC for mild cases (98 eyes), 0.83–0.84 for moderate cases (180 eyes) and 0.95–1.00 for severe non-proliferative and proliferative cases, which are few (30 severe non-proliferative eyes in the validation set).

4.3. Export fidelity and edge performance

Table 4 reports the final 512×512 model on the held-out test set (1025 images) on both devices. Latency and memory depend on the architecture and input size, not on the trained weights, so Table 5 compares 224×224 and 512×512 inputs with the same protocol.

Table 4

Export fidelity of the final 512×512 model on the held-out test set (1025 images). AUCs are per label, with DR scored against per-eye labels. $|\Delta p|_{\max}$ is the maximum absolute probability difference from the PyTorch CPU reference.

Device	Runtime	DR	Glaucoma	Cataract	AMD	HR	$ \Delta p _{\max}$
Workstation	PyTorch FP32 (reference)	0.8093	0.8916	0.8973	0.8616	0.8126	–
N100	ONNX Runtime FP32, CPU	0.8093	0.8916	0.8973	0.8616	0.8126	2.5×10^{-5}
N100	OpenVINO FP32, CPU	0.8093	0.8916	0.8973	0.8616	0.8126	1.0×10^{-5}
N100	OpenVINO, iGPU	0.8083	0.8913	0.8979	0.8625	0.8129	6.4×10^{-2}
Pi 3B	TFLite FP32, 4 threads	0.8093	0.8916	0.8973	0.8616	0.8126	1.3×10^{-4}

Export of the final model is lossless on every CPU runtime: all five test AUCs equal the PyTorch reference to four decimals, and per-image probabilities differ by at most 1.3×10^{-4} . The N100’s integrated GPU runs in reduced precision by default. It shifts probabilities by up to 0.064 and AUCs by up to 0.001, so we treat CPU inference as the reference deployment.

At 512×512 the N100 takes 7–8 ms per image for the model and about 14–16 ms end to end. At this resolution the integrated GPU is the fastest backend, the reverse of the 224×224 case. The Pi 3B takes 0.51 s for the model and about 0.74 s end to end (1.35 images per second), and peaks at 65 MB. Free memory never fell below 666 MB, so the 870 MB limit is not binding even at full resolution. Throughout, the Pi reported under-voltage throttling, with the clock at 600 MHz in every sample during the four-thread runs. A correctly powered board should therefore be roughly twice as fast.

Table 5

Latency and peak memory at 224×224 and 512×512 input (FP32, batch size 1). Model-only latency is the mean over timed runs; end-to-end adds JPEG decoding, resizing and normalisation. The Pi was under-voltage throttled to 600 MHz during these runs.

Device	Runtime	Model-only (ms)		End-to-end (ms) 224 / 512	Peak RSS (MB) 224 / 512
		224	512		
N100	ONNX Runtime, CPU	2.06	8.29	7.3 / 15.9	196 / 222
N100	OpenVINO, CPU	2.43	8.40	7.0 / 16.1	238 / 257
N100	OpenVINO, iGPU	3.03	6.93	7.4 / 13.9	613 / 617
Pi 3B	TFLite, 4 threads	130.4	512.5	224 / 739	48 / 65
Pi 3B	TFLite, 1 thread	172.6	856.6	263 / 1079	44 / 65

Development model at 224×224 , including INT8. Table 6 reports the same benchmark for the 224×224 development model on the 1270-image validation set, including INT8 variants. The INT8 calibration images were drawn from the original training split, and 55 of them later fell into the held-out test set. INT8 is not evaluated on the test set, so no test-set number is affected.

Table 6

Edge benchmark of the 224×224 model on the 1270-image validation set. Latency is model-only at batch size 1 (mean / 95th percentile). $|\Delta p|_{\max}$ is the maximum absolute probability difference from the PyTorch reference. Pi runs were power-capped at 600 MHz (see text).

Device	Configuration	Mean AUC	DR AUC	$ \Delta p _{\max}$	Latency (ms)	RSS (MB)
N100	ONNX Runtime FP32, CPU	0.8214	0.6947	6.3×10^{-6}	2.06 / 2.05	196
	OpenVINO FP32, CPU	0.8215	0.6947	6.2×10^{-6}	2.43 / 2.44	238
	OpenVINO INT8, CPU	0.8050	0.6673	0.92	1.97 / 1.98	242
	OpenVINO FP32, iGPU	0.8215	0.6944	0.059	3.03 / 3.08	613
	OpenVINO INT8, iGPU [†]	0.6226	0.4650	0.999	3.24 / 3.31	657
	OpenVINO INT8, iGPU, fixed [‡]	0.8037	0.6648	0.93	2.83 / 2.87	–
Pi 3B	TFLite FP32, 4 threads	0.8215	0.6947	1.8×10^{-5}	130.4 / 140.3	48
	TFLite FP32, 1 thread	0.8214	0.6947	1.7×10^{-5}	172.6 / 175.9	44
	TFLite dyn.-range INT8, 4 thr.	0.8166	0.6819	0.70	165.4 / 195.9	39
	TFLite dyn.-range INT8, 1 thr.	0.8166	0.6818	0.70	165.1 / 169.3	41
	TFLite full INT8, 4 threads	0.5926	0.4944	0.98	78.3 / 78.5	38
	TFLite full INT8, 1 thread	0.5861	0.4963	0.98	151.1 / 162.5	38

[†]The OpenVINO GPU plugin returns NaN in INT8 depthwise convolutions; the result is identical with Intel's current GPU driver, so it is not a driver fault (Section 4.4). [‡]Depthwise convolutions excluded from quantization; measured with Intel compute-runtime

26.35; memory not recorded.

FP32 export is lossless on both devices: on all 1270 images, the outputs match the reference to within 2×10^{-5} in probability, and the AUCs are identical to the third decimal. On the N100 the model takes about 2 ms, so preprocessing in Python (about 5 ms) dominates end-to-end time, which stays well above 100 images per second. The integrated GPU is slower than the CPU at batch size 1 and uses about three times the memory. Replacing the 2022 Debian GPU driver with Intel's current compute runtime (26.35) shortened iGPU latency by only 9% (3.02 to 2.75 ms), and left it unchanged within 1% at 512×512 . The gap to the CPU therefore reflects per-inference dispatch overhead on a very small model, not a driver fault. On the Pi 3B the FP32 model takes 130 ms with four threads, plus about 90 ms of preprocessing, and peaks at 48 MB, a small fraction of the 870 MB available. The Pi reported under-voltage throttling throughout (`get_throttled=0x50005`, 600 MHz), so its latencies are pessimistic relative to a correctly powered board.

4.4. INT8 quantization

On the N100, naive NNCF INT8 saved under 0.5 ms per image and cost 0.017 mean AUC: DR dropped by 0.027, glaucoma by 0.030 and cataract by 0.024. Per-image probabilities shifted by up to 0.92, which would also invalidate

any downstream confidence calibration. On the Pi, dynamic-range quantization cost 0.005 AUC and was *slower* than FP32 with four threads. Full-integer quantization was $1.7\times$ faster but collapsed DR to chance (AUC 0.49). This failure is reproducible on the workstation and on training images. It persisted when the squeeze-and-excitation pooling and the most-affected layers were kept in floating point, which indicates error accumulated across the network rather than a single faulty layer. On the N100's integrated GPU, the INT8 model collapsed to a mean AUC of 0.62 although the same model reached 0.805 on the CPU. The collapse was identical with the 2022 Debian GPU driver and with Intel's current compute runtime, and with 16-bit or 32-bit inference precision, so neither the driver nor reduced precision causes it. Re-quantizing with other NNCF presets or with the GPU as target device did not help. Comparing every intermediate output between CPU and GPU located the fault in the first depthwise convolution, whose GPU output contained NaN in two of its sixteen channels on every image tested. A single-layer reproduction isolated the trigger. NNCF had quantized the depthwise weights with one range per output channel *and* per kernel column, rather than the usual one per channel. This layout is valid and the CPU computes it correctly, but the GPU plugin produced about 5% error in every channel and NaN in the last two. Collapsing the ranges to one per channel removed the error entirely. Excluding depthwise convolutions from quantization restored INT8 accuracy on the iGPU (mean AUC 0.804, equal to CPU INT8). At 2.83 ms, however, the repaired model is no faster than the iGPU in its default 16-bit mode, and it keeps INT8's loss of accuracy and calibration. Because the FP32 model already fits comfortably within the Pi's memory, we recommend FP32 deployment for this model class.

4.5. Grad-CAM evaluation

Fidelity and cost. On 300 test images and all five labels, the edge implementation (exported backbone plus NumPy head) reproduced the PyTorch autograd maps to within 2.3×10^{-4} (ONNX) and 7.2×10^{-4} (TensorFlow Lite) on a 0–1 scale. The location of the map maximum was identical in 1499 of 1500 maps. On the devices themselves the maps stayed within 2.5×10^{-4} (N100) and 3.9×10^{-4} (Pi) of the reference. Producing a DR map, or maps for all five labels, added 1.7 and 4.9 ms to the N100's 7.5 ms inference (ONNX Runtime). On the power-capped Pi it added 50 and 146 ms to 547 ms (9% and 27%). Peak memory grew by at most 19 MB, to 81 MB on the Pi.

Sanity check. Grad-CAM passed the model-parameter randomisation test [54]. Re-initialising only the classifier dropped the median Spearman correlation with the original maps to 0.075, below the 0.144 between maps of two unrelated images. Randomising deeper layers kept it between -0.045 and 0.022 . Maps of different labels for the same image were nearly unrelated (median correlation 0.087), so the maps are class-specific. The data-randomisation variant was not run.

Faithfulness. Over all positive test pairs, inserting pixels in Grad-CAM order recovered the prediction much faster than random or centre-first order. Insertion AUC was 0.740, against 0.471 (random) and 0.593 (centre), and the margin over centre was 0.147 (95% CI 0.134 to 0.160). Deletion separated Grad-CAM from random (0.326 vs 0.467) but not from the centre prior (0.341, $p = 0.20$). For DR alone, deletion was *worse* than the centre prior (0.331 vs 0.294). For cataract, deletion did not beat random order, which fits a diffuse lens opacity better than a focal lesion.

Table 7

Localisation of the DR Grad-CAM against IDRiD expert lesion masks (81 images; soft exudates present in 40). Pixel AUROC is averaged over images; the pointing game is the fraction of images whose map maximum lies inside a lesion of that type.

Lesion type	Grad-CAM (DR)	Pixel AUROC		Glaucoma map	Pointing game Grad-CAM / random / centre
		Random	Centre		
Any lesion	0.657	0.499	0.575	0.518	0.14 / 0.04 / 0.09
Haemorrhages	0.708	0.502	0.477	0.505	0.10 / 0.02 / 0.01
Hard exudates	0.640	0.499	0.711	0.535	0.02 / 0.01 / 0.06
Soft exudates	0.569	0.502	0.397	0.498	0.00 / 0.00 / 0.03
Microaneurysms	0.622	0.499	0.580	0.515	0.01 / 0.00 / 0.00

Localisation. On external IDRiD images (Table 7), the DR map overlapped expert lesions better than the random, centre and wrong-class baselines (pixel AUROC 0.657 for any lesion). It placed 1.7 times more of its mass inside

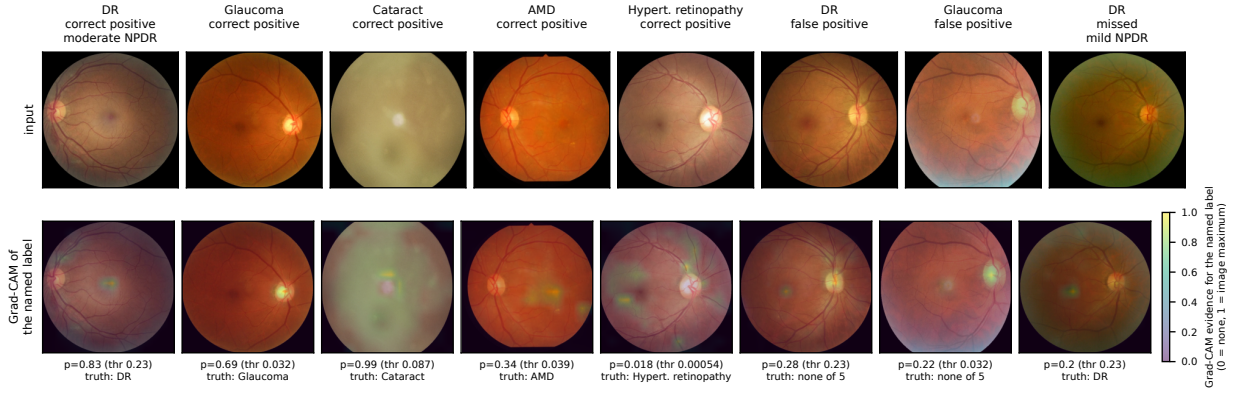


Figure 1: Grad-CAM of the deployed model on held-out test images: one correct positive per label, a DR and a glaucoma false positive, and a missed mild non-proliferative DR case. Each example is a seeded random draw from its category. Colour shows positive evidence for the named label, scaled to the image maximum. Below each panel: predicted probability, validation-chosen threshold and ground truth.

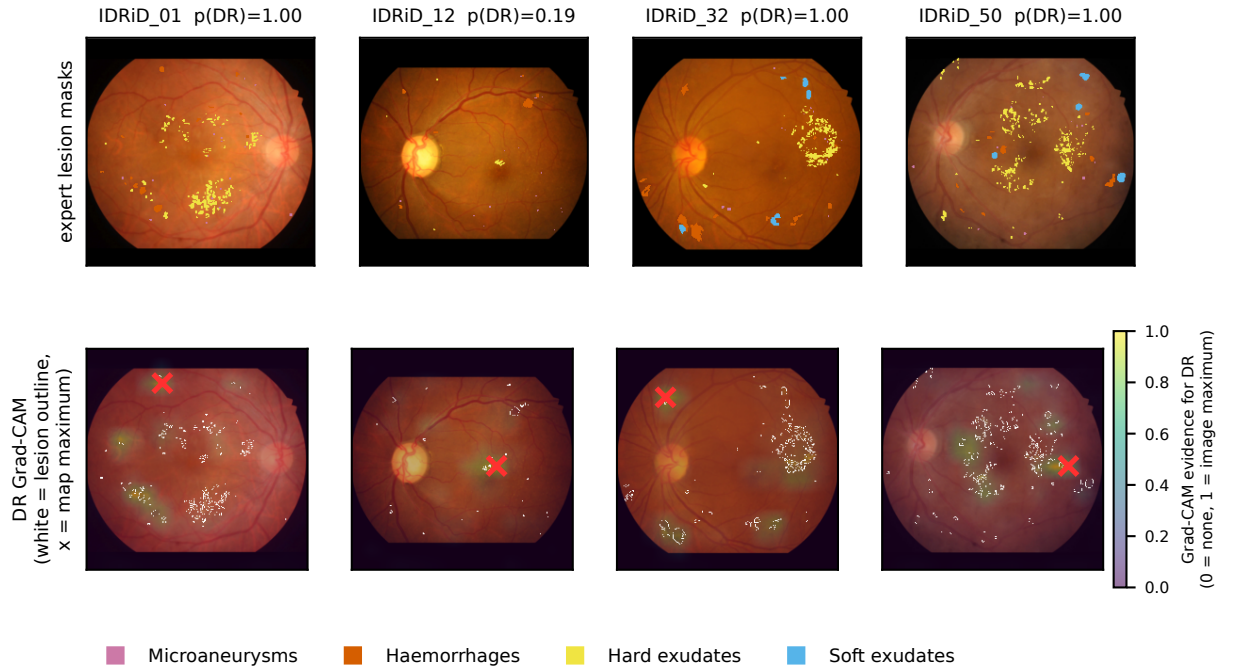


Figure 2: IDRiD images with expert lesion masks (top) and the DR Grad-CAM with lesion outlines in white and the map maximum marked (bottom).

lesions than their area alone would predict. Agreement was best for haemorrhages. For hard exudates, which cluster around the macula near the image centre, the centre prior did better. The map maximum fell on a microaneurysm in only 1 of 81 images. At the network's 16×16 feature resolution, lesions a few pixels wide cannot be pointed at, which is consistent with saliency benchmarks on small pathologies [57]. Fig. 1 shows randomly drawn examples per category, including failures, and Fig. 2 shows IDRiD overlays. On ODIR several maps peak at the optic disc or the fovea rather than at lesions.

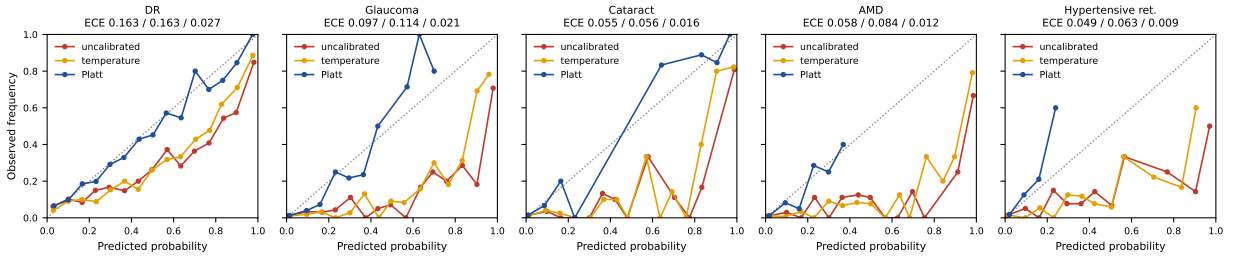


Figure 3: Reliability diagrams of the deployed model on the held-out test set: uncalibrated outputs, temperature scaling and Platt scaling, each fitted on the validation set. Titles give ECE for the three variants in that order. Bins with fewer than five images are not drawn.

4.6. Confidence calibration

Table 8 and Fig. 3 show calibration of the deployed model on the test set. Raw outputs overestimate disease probability: the mean predicted probability is 1.5–2.6 times the prevalence, consistent with the positive-class weights used in training. Temperature scaling did not help. The temperatures that minimised validation NLL were all above one (1.12–1.58). They pushed probabilities towards 0.5 and left ECE unchanged for DR and cataract, and worse for glaucoma, AMD and hypertensive retinopathy. Platt scaling reduced ECE for every label, by 0.04 to 0.14 with confidence intervals of the reduction excluding zero. It brought the mean predicted probability to within 0.01 of the prevalence. The same pattern held for all nine trained models (ECE after Platt scaling 0.009–0.034, averaged over seeds).

Table 8

Calibration of the deployed model on the held-out test set. ECE uses 15 equal-width bins. Δ is the change in ECE from Platt scaling with its 95% patient-bootstrap CI.

Label	Prevalence	ECE raw	ECE temperature	ECE Platt	Δ ECE (95% CI)
DR (per-eye)	0.278	0.163	0.163	0.027	−0.136 [−0.143, −0.094]
Glaucoma	0.060	0.097	0.114	0.021	−0.076 [−0.086, −0.054]
Cataract	0.060	0.055	0.056	0.016	−0.039 [−0.049, −0.024]
AMD	0.041	0.058	0.084	0.012	−0.046 [−0.055, −0.028]
Hypertensive ret.	0.033	0.049	0.063	0.009	−0.040 [−0.054, −0.018]

Calibrated outputs survive deployment. Applying the validation-fitted Platt maps to the probabilities computed on the N100 CPU runtimes and on the Pi gives the same ECE as the reference to four decimals for every label, with calibrated probabilities within 1.4×10^{-4} of it. The reduced-precision integrated GPU changes DR ECE from 0.027 to 0.028. Quantization is different. On the 224×224 development model, a Platt map fitted to FP32 outputs and applied to full-integer INT8 outputs of the same validation images roughly tripled DR ECE (0.025 to 0.080) and nearly quadrupled cataract ECE (0.014 to 0.051). Dynamic-range INT8 roughly preserved calibration.

4.7. Energy

Table 9 reports N100 package energy measured with Intel RAPL over sustained 60 s loops (three repeats, mean $\pm$ s.d.). Idle package power was 2.6 W.

At 512×512 the integrated GPU needs 42–45% less energy per inference than either CPU backend. Under sustained load the N100 settles at its 10 W long-term power limit, and the package reached 80 °C. Sustained CPU throughput at 512×512 (about 10.5 ms per inference) is therefore lower than the single-shot latency in Table 5 suggests. ONNX Runtime’s default thread pool busy-waits on all four cores, and its power figures include this. Even end to end at full resolution, one screening image costs about 0.13–0.20 J, negligible next to the camera and display. The Pi has no on-board energy counter and was not measured.

Table 9

N100 energy per inference (package domain, RAPL). Above-idle subtracts idle package power. Model-only loops repeat inference on one preprocessed image; end-to-end loops decode and preprocess each image.

Input	Runtime	Power (W)	Inferences/s	mJ / inference	mJ above idle
<i>Model-only</i>					
224	ONNX Runtime, CPU	11.32 ± 0.17	428.5 ± 16.2	26.4 ± 0.6	20.3 ± 0.4
224	OpenVINO, CPU	11.31 ± 0.03	346.2 ± 2.4	32.7 ± 0.1	25.1 ± 0.1
224	OpenVINO, iGPU	9.31 ± 0.02	333.1 ± 0.9	28.0 ± 0.1	20.1 ± 0.1
512	ONNX Runtime, CPU	11.53 ± 0.21	95.6 ± 3.8	120.7 ± 2.5	93.6 ± 1.4
512	OpenVINO, CPU	11.50 ± 0.01	90.1 ± 0.3	127.7 ± 0.3	99.0 ± 0.3
512	OpenVINO, iGPU	10.07 ± 0.01	144.6 ± 0.2	69.6 ± 0.2	51.7 ± 0.2
<i>End-to-end</i>					
224	ONNX Runtime, CPU	9.70	139.3	69.6	50.9
224	OpenVINO, CPU	10.66	141.6	75.3	56.8
224	OpenVINO, iGPU	9.00	128.3	70.2	49.8
512	ONNX Runtime, CPU	10.99	59.1	186.1	142.3
512	OpenVINO, CPU	11.08	54.6	202.9	155.5
512	OpenVINO, iGPU	9.81	74.0	132.6	97.6

5. Discussion

Both tiers are sufficient for throughput. A screening camera produces at most a few images per minute. Both devices run the model far faster than that: the N100 at about 60–140 images per second end to end, and even the power-capped Pi 3B at 1.35 images per second at 512×512 . The cheaper tier is therefore not limited by speed for single-camera screening, and with 65 MB peak memory at full resolution it is not limited by memory either. The binding constraint is accuracy.

Quantization is not needed and naive quantization is harmful. INT8 is usually justified by memory and speed. Here the FP32 model peaks at 48 MB on the Pi, a small fraction of its memory, and INT8 bought little or no speed on either device. At the same time, naive post-training quantization reduced accuracy and shifted per-image probabilities by up to 0.92 (OpenVINO) and 0.98 (TensorFlow Lite full-integer), which would invalidate any calibrated confidence reported to a health worker. This agrees with documented post-training quantization difficulties of depthwise-separable networks [32, 33] and contrasts with edge reports that describe INT8 as nearly free [38]. In our TensorFlow Lite experiments the error was spread through the network rather than confined to the SE blocks we inspected. Quantization-aware training or bias-corrected quantization [31, 34] may recover accuracy; we did not test them. Our recommendation is to deploy FP32 for this model class unless memory forces otherwise.

Calibration needs a bias term. Weighting the loss by inverse prevalence is the usual remedy for rare labels, but it leaves every output biased towards disease. Temperature scaling is often presented as a near-universal fix [50], yet it cannot remove a constant logit offset and made calibration worse here. A two-parameter Platt map removed the offset at negligible cost. Any screening tool that shows confidence to a non-specialist and is trained with class weighting should check for this offset. Because full-integer INT8 shifts probabilities enough to invalidate a map fitted in floating point, the calibration step and the deployed numeric format must be validated together.

DR remains the weakest label. At 512×512 the deployed model's DR AUC (0.78 averaged over seeds, 0.81 for the deployed seed) stays below the 0.93–0.99 reported for systems trained on large, adjudicated datasets [5, 15]. It is closer to a reproduction on public data with single-grader labels [16]. Referable DR is detected better (AUC 0.87), and most residual errors are in mild non-proliferative DR, whose microaneurysms span only a few pixels even at 512 pixels. Per-eye labels removed an obvious source of label noise (18.9% of DR-positive images show no DR in that eye) but did not improve test AUC. Label quality therefore seems to limit performance less than image detail does. The preprocessed ODIR-5K images are limited to 512×512 , so higher input resolution would require the original images.

Limitations. All training and testing use a single dataset, ODIR-5K, whose images were preprocessed by a third party, and there is no external validation of classification performance. Candidate external sets are RFMiD [25] and Messidor [58]; IDRiD [55] was used only for lesion localisation. Resolution and the DR label definition were chosen on the validation set, which also selects checkpoints and fits calibration maps. The held-out test set guards the reported numbers but not these design choices. Test sets are small for the rarer labels (34–62 positives), as the wide confidence intervals show. The Pi 3B was power-capped throughout, so its latencies are pessimistic. Each latency configuration was measured once, and the Pi's energy was not measured. INT8 was evaluated only on the 224×224 development model. Grad-CAM was evaluated computationally but not in a reader study, and published evidence cautions that saliency maps can mislead readers [57, 59, 60].

6. Conclusion

A single MobileNetV3-small network screened five retinal conditions from one fundus photograph. It ran on a mini PC and on a Raspberry Pi 3B with under 1 GB of RAM with no measurable loss of accuracy in floating point. On the Pi the full 512×512 pipeline, including a calibrated output and a Grad-CAM map, takes under one second per image and stays below 100 MB of memory. Three lessons generalise beyond this model. First, input resolution mattered more for DR than cleaning the labels. Second, naive INT8 quantization was not worth its cost, because it degraded both accuracy and calibration while memory was not a constraint. Third, a network trained with class weights needs a calibration map with a bias term before its probabilities can be shown as confidence. The remaining gaps are clinical: mild DR is poorly detected, Grad-CAM cannot show microaneurysms, and the model has not been validated on an external population or with health workers. Those are the next steps.

CRedit authorship contribution statement

Shalini Agarwal: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft. **Aruna Bhat:** Conceptualization, Supervision, Validation, Writing – review & editing. [TODO: Authors to confirm the role split.]

Declaration of competing interest

The authors declare that they have no conflict of interest.

Data availability

ODIR-5K [17, 18] and IDRiD [55] are publicly available. [TODO: Code, export scripts, benchmark harness and trained weights: link to public repository / Zenodo DOI.]

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors. All analyses use publicly available, de-identified datasets.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to help write and run analysis and benchmarking code and to draft and edit the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. [TODO: Authors to confirm the wording matches actual use.]

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